

Syllabus for Computer Logic and Circuit Design: PHYS306/COSC330

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August 11, 2026

Abstract

The core concepts of digital logic and digital circuit design are presented via integrated lectures and laboratory activities. The course begins with binary mathematics and floating point representation. The concepts of Boolean logic and logic gates, truth tables, combinatorial logic and Karnaugh maps follow. These core concepts prove useful in engineering, business, and science. Hands-on experience with programmable logic (firmware) and electronics lab equipment are emphasized in lab activities. Following the core modules, more complex digital functions are introduced. Examples include adders, comparators, multiplexers and demultiplexers, latches and flip-flops (memory elements), registers and shift registers, counters, and finite-state-machines (FSMs). We conclude with analog-to-digital conversion (ADCs), and digital-to-analog conversion (DACs). The ADC/DAC module builds to digital signal processing (DSP), as time permits. Students will create and present final projects that integrate course concepts using programmable logic and Python code.

Pre-requisites: *PHYS180, COSC120. Pre-requisites may be waived at instructor discretion.*

Course credits, Liberal Arts Categorization: *3 Credits, QR2*

Regular course hours: *Tuesdays and Thursdays from 13:30 - 14:50 in SLC 104.*

Instructor contact information: *Discord: 918particle, Email: jhanson2@whittier.edu, Office: SLC 222, Book appointments: <https://calendar.app.google/gYfm8uzRbZ7fW9Dc7>.*

Office hours: *Use booking link above to schedule office hours.*

Attendance/Absence: *Students needing to reschedule midterms must notify the professor a few days in advance.*

Late work policy: *Late work is generally not accepted, but is left to the discretion of the instructor.*

Text: *Digital Fundamentals, Eleventh Edition., by Thomas L. Floyd (Pearson). This is a widely adopted text with a variety of online features, bonus chapters, code examples, and excellent homework sets. Students can borrow editions 7-9 from the Dept. of Physics and Astronomy, or purchase the 11th edition.*

Grading: *The course grade will be a weighted average of assignment scores, and the weights are listed in Tab. 1.*

Assignment	Weight	Date
Homework sets	30%	Weekly on Fridays
Midterm 1	25%	Thursday, October 1st
Midterm 2	25%	Thursday, November 5th
Final Project	20%	December 1st and 3rd

Table 1: A list of major assignments and grade weights. The midterm with the higher score will be weighted 30%, and the midterm with the lower score will be weighted 25%.

Grade Settings: $< 60\% = F$, $> 60\%, \leq 70\% = D$, $> 70\%, \leq 80\% = C$, $> 80\%, \leq 90\% = B$, $< 90\%, \leq 100\% = A$.
Pluses and minuses: 0-3% minus, 3%-6% straight, 6%-10% plus (e.g. 79% = C+, 91% = A-)

Homework Sets: *Due weekly on Fridays, including textbook problems, submitted code, and digital designs.*

Student Accessibility Services (SAS) statement: *Whittier College is committed to making learning experiences as accessible as possible. If you experience physical or academic barriers due to a disability, you are encouraged to contact Student Accessibility Services (SAS) to discuss options. To learn more about academic accommodations, email sas@whittier.edu, call (562) 907-4825, or visit the SAS office on the ground floor of Wardman Library.*

Academic Honesty Policy: http://www.whittier.edu/academics/academic_honesty

Course Objectives:

- Fluency with number systems and binary codes, logic gates and Boolean algebra, and complex logic functions
- Application of digital logic in science, engineering, and business administration
- Practice translating a diagram to a prototype, and troubleshooting
- Create and control programmable logic using Python
- Create and verify final project designs that accomplish design tasks
- Present final project results to peers

Course Outline:

1. Week 0 - Reading: Chapter 1 of Digital Fundamentals.

- (a) First meeting - Thursday, August 20th, 2026.
 - i. Course introduction, syllabus distribution
 - ii. Digital concepts lecture content, block diagram of designs
 - iii. Digital concepts and introduction of integrated circuits
 - iv. *Laboratory tour and introductory laboratory exercises*
 - A. Introduction to system on a chip (SoC): PYNQ-Z1
 - B. Basic Linux terminal commands
 - C. The PYNQ-Z1 SoC and Jupyter Notebooks
 - D. Oscilloscopes and signal generation

2. Week 1 - Reading: Chapters 1-2 of Digital Fundamentals.

- (a) First meeting - Tuesday, August 25th, 2026. *Reading: Chapter 1.*
 - i. Logic functions: basic, combinatorial, and serial and parallel
 - ii. Programmable logic and fixed IC logic
 - iii. Test and measurement instruments
 - iv. *Laboratory exercise, part I*
 - A. Digital signals from PYNQ-Z1
 - B. PL layer and `logictools` library
- (b) Second meeting - Thursday, August 27th, 2026. *Reading: Chapters 2-1 through 2-7*
 - i. Decimal and binary numbers, conversions and arithmetic, signed numbers
 - ii. *Laboratory exercise, part II*
 - A. Representing a binary number with LEDs and `logictools`
 - B. Capturing digital signals with the oscilloscope

3. Week 2 - Reading: Chapter 2 of Digital Fundamentals.

- (a) First meeting - Tuesday, September 1st, 2026. *Reading: Chapter 2-1 through 2-7.*
 - i. Binary arithmetic, signed numbers
 - ii. *Laboratory exercise, part I*
 - A. Representing a binary number with LEDs, part II
 - B. The `logictools` boolean generator: multiple logic functions, associative property
- (b) Second meeting - Thursday, September 3rd, 2026. *Reading: Chapters 2-8, 2-10 through 2-12*
 - i. Hexadecimal numbers
 - ii. BCD, gray codes, and the parity and CRC error codes
 - iii. *Laboratory exercise, part II*
 - A. Breadboard prototyping
 - B. Digital voltmeters, DC power supplies

4. Week 3 - Reading: Chapter 3 of Digital Fundamentals.

- (a) First meeting - Tuesday, September 8th, 2026. *Reading: Chapters 3-1 through 3-3*
 - i. Hexadecimal numbers
 - ii. BCD, gray codes, and the parity and CRC error codes
 - iii. *Laboratory exercise, part I*
 - A. Decimal, binary, and hexadecimal conversion (Python codelab)
 - B. Random number generation, and matplotlib (`common` Jupyter notebooks)
 - C. PYNQ-Z1 and `logictools`: finite state machines (FSMs) and logic simplification, part I
- (b) Second meeting - Thursday, September 10th, 2026. *Reading: Chapters 3-4 through 3-6*
 - i. Logic gates: NOT, AND, OR, NAND, NOR, XOR, XNOR
 - ii. *Laboratory exercise, part II*
 - A. PYNQ-Z1 and `logictools`: finite state machines (FSMs) and logic simplification, part II

5. Week 4 - **Reading: Chapters 3-4 of Digital Fundamentals.**

- (a) First meeting - Tuesday, September 15th, 2026. *Reading: Chapters 3-8 and 3-9, 4-1 and 4-2*
 - i. Programmable logic with gates
 - ii. Boolean algebra, with applications and examples in business and engineering
 - iii. *Laboratory exercise, part I*
 - A. PYNQ-Z1 and **logictools**: finite state machines (FSMs) and logic simplification, part III
- (b) Second meeting - Thursday, September 17th, 2026. *Reading: Chapters 4-3 through 4-10*
 - i. DeMorgan's Theorems; further abstract examples in business and engineering
 - ii. Boolean analysis of logic circuits, simplification
 - iii. Truth tables, Karnaugh maps and SOP/POS minimization
 - iv. *Laboratory exercise, part II*
 - A. PYNQ-Z1 **logictools**: trace analyzer and pattern generator
 - B. Use pattern generator to create multi-pin digital patterns, view with trace analyzer

6. Week 5 - **Reading: Chapter 5 of Digital Fundamentals**

- (a) First meeting - Tuesday, September 22nd, 2026. *Reading: Chapters 5-1 through 5-3*
 - i. Combinatorial systems of logic gates
 - ii. Universal NAND and NOR gates
 - iii. *Laboratory exercise, part I*
 - A. Plot the trace of combinatorial logic, part I
- (b) Second meeting - Thursday, September 24th, 2026. *Reading: Chapters 5-4 through 5-7*
 - i. *Laboratory exercise, part II*
 - A. Cumulative exercise with boolean generator, pattern generator, trace analyzer, and stepper

7. Week 6 - **Reading: Review chapters 1-5 of Digital Fundamentals**

- (a) First meeting - Tuesday, September 29th, 2026.
 - i. Review material, collection of example problems
 - ii. Catch-up on laboratory exercises and troubleshooting
- (b) Second meeting - Thursday, October 1st, 2026.
 - i. **First midterm exam**
 - A. Covers material from chapters 1-5
 - B. Emphasis on binary operations, boolean algebra, and simple collections of gates

8. Week 7 - **Reading: Chapter 6 of Digital Fundamentals**

- (a) First meeting - Tuesday, October 6th, 2026. *Reading: Chapters 6-1 through 6-6*
 - i. Adders, comparators, and encoders
 - ii. *Laboratory exercise, part I*
 - A. Probing fixed IC gates with scope and LEDs
- (b) Second meeting - Thursday, October 8th, 2026. *Reading: Chapters 6-7 through 6-11*
 - i. Multiplexers and Demultiplexers
 - ii. *Laboratory exercise, part II*
 - A. Comparing fixed IC gates to programmable logic

9. Week 8 - **Reading: Chapters 7 and 8**

- (a) First meeting - Tuesday, October 13th, 2026. *Reading: Chapters 7-1 through 7-4, 7-5*
 - i. S-R latches
 - ii. Flip-flops, D and J/K
 - iii. *Laboratory exercise, part I*
 - A. Probing fixed IC counters with scope and LEDs, part I
 - B. Final project development
- (b) Second meeting - Thursday, October 15th, 2026. *Reading: Chapters 7-5, and 8-1 through 8-4*
 - i. One-shots (timers) and pulse generation
 - ii. Shift registers
 - iii. *Laboratory exercise, part II*

- A. Probing fixed IC counters with scope and LEDs, part II
- B. Final project development

10. Week 9 - **Reading: Chapters 8 and 9**

- (a) First meeting - Tuesday, October 20th, 2026. *Reading: Chapters 8-5, 9-1 through 9-5*
 - i. Shift register applications
 - ii. Finite state machines (FSMs)
 - iii. Asynchronous and synchronous counters
 - iv. *Laboratory exercise, part I*
 - A. Catch-up on laboratory exercises and troubleshooting
 - B. *Work on final project proposals*
 - C. Optional: FSM and counter example with `logictools`
- (b) Second meeting - Thursday, October 22nd, 2026. *Reading: 9-6 through 9-8*
 - i. Cascaded counters and applications of counters
 - ii. *Laboratory exercise, part II*
 - A. Study period for second midterm

11. Week 10 - **Reading: Review chapters 6-9**

- (a) First meeting - Tuesday, October 27th, 2026.
 - i. Review chapters 6-9
 - A. Adders, encoders/decoders, multiplexing
 - B. Latches, flip-flops, timers and pulse generation
 - C. Shift registers, counters, finite-state machines (FSMs)
 - ii. Review of internship programs and research with the US Navy
- (b) Second meeting - Thursday, October 29th, 2026. *Reading: Chapters 12-1 through 12-3*
 - i. Analog-to-Digital Conversion (ADC)
 - ii. Digital-to-Analog Conversion (DAC)
 - iii. *Laboratory exercise, part I*
 - A. ADC/DAC with PMODs, part I
 - B. Produce analog voltage from programmable logic, read analog voltage to programmable logic

12. Week 11 - **Reading: Chapter 12**

- (a) First meeting - Tuesday, November 3rd, 2026. *Reading: Chapters 12-4 through 12-5*
 - i. Digital Signal Processing (DSP) - an introduction to sampling and digitization
 - ii. *Laboratory exercise, part II*
 - A. ADC/DAC with PMODs, part II
 - B. ADC/DAC and Fourier series
- (b) Second meeting - Thursday, November 5th, 2026.
 - i. **Second midterm exam**
 - A. Covers material from chapters 6-12
 - B. Emphasis on complex logic functions, memory elements and registers, FSMs, and ADC/DAC

13. Week 12 - **Reading: Chapters 2-3 of Digital Signal Processing**

- (a) First meeting - Tuesday, November 10th, 2026. *Reading: Chapter 2 of DSP*
 - i. Selected topics in statistics and probability
 - ii. Mathematics of sampling and digitization
 - iii. *Laboratory exercise, part I*
 - A. Hardware acceleration of an FIR data filter, part I
 - B. Creation of custom PYNQ firmware with open-source IP
- (b) Second meeting - Thursday, November 12th, 2026. *Reading: Chapter 3 of DSP*
 - i. The sampling theorem
 - ii. Aliasing and anti-aliasing
 - iii. *Laboratory exercise, part II*
 - A. Hardware acceleration of an FIR data filter, part II

B. Measurement of the speed-up factor during data filtering

14. Week 13 [*Optional extra topics*] - **Reading: Chapters 30-31 of Digital Signal Processing**

(a) First meeting - Tuesday, November 17th, 2026. *Reading: Chapter 30 of DSP Guide*

- i. Complex numbers and the usage of complex numbers in DSP
- ii. *Laboratory exercise, part I*
 - A. Exploring the Fourier transform, part I
 - B. Power spectra and spectrograms

(b) Second meeting - Thursday, November 19th, 2026. *Reading: Chapter 31 of DSP Guide*

- i. Complex numbers and the FFT, filters
- ii. *Laboratory exercise, part II*
 - A. Exploring the Fourier transform, part II
 - B. Filters
 - C. Hardware-accelerated FIR filter and power spectrum

15. Week 14 - **Fall/Thanksgiving Break:** November 23rd–27th, 2026. No instruction.

16. Week 15 - **Group presentations and Final Exam**

(a) Meetings - Tuesday, December 1st, 2026, and Thursday, December 3rd, 2026.

- i. *Group presentations: 15 minutes each, plus time for questions*
- ii. *Final presentations are given in person.*